



Department of Computer Science and Business Systems
Academic Year 2023 – 2024(Odd Semester)

Degree, Semester & Branch: B.Tech., I & CSBS

Course Code & Title: GE3151 Problem Solving and Python Programming

Name of the Faculty member: Ms.K.Usharani AP/CSBS

Innovative Practice Description

- **Unit / Topic:** Unit 4 - Dictionaries: operations, Methods
- **Course Outcome:** CO4– Use Python data structures - lists, tuples, dictionaries to represent complex data.
- **Topic Learning Outcome:** TLO 12 -Students will be able to apply the various functions of list, tuples, dictionary data structure for solving real time problems
- **Activity Chosen:** Cross Word Puzzle
- **Justification:**
To identify the understanding level of students on the methods of dictionary, list and tuple.
- **Time Allotted for the Activity:** 15 Minutes
- **Details of the Implementation:**
 1. The activity was conducted in the class room with 62 number of students.
 2. The announcement was given to the students in-prior.
 3. In a class room, a printed sheet of 10 - 12 questions is given to each student. The students are asked to fill the answers in the given sheet and submit back to me.
 4. The sheets were distributed randomly to students for correction.
 5. The answers were discussed in the class room immediately to make them to know answer for all questions.
- **CO – PO / PSO mapping:**

CO	PO1	PO2	PO3	PO12	PSO2
CO2	3	2	1	1	2

(1 – Low 2 – Moderate 3 – High)

- **PO / PSO mapped:**

Innovative practice	PO1	PO2	PO3	PO12	PSO2
	3	2	1	1	2
Justification for correlation	Apply the knowledge of	The student should identify the	Design solutions for complex	Recognize the need and have ability to	Exhibit the proficiency in software

	engineering fundamentals	basic knowledge on methods of dictionary	engineering problems	engage in independent and life-long learning.	development
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• Images / Screenshot of the practice:

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ACROSS

- Function that deletes all entries and returns {} ✓
- del(1, 2, 3, 4) ✓
- Function that returns x/3 ✓
- Keys in the dictionary must be of any mutable data type ✓
- del(1, 2, 3, 4) ✓
- Function that returns (1, 2, 3, 4) ✓
- Tuples can be used as key for dictionary (True/False) ✓
- del(1, 2, 3, 4) ✓

DOWN

- Function that constructs a list from those elements of the list for which a function returns true ✓
- print(1, 2, 3, 4) ✓
- Lists can be used as key for dictionary (True/False) ✓
- When a list is assigned to another using the assignment operator, then a new copy of the list is made ✓
- Which data structure allows you to return multiple values from a function? ✓
- It is impossible to add, delete elements from a tuple ✓

11/12
A. Dharshant

G. Santhosh

ACROSS

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- del(1, 2, 3, 4) ✓
- Function that returns x/3 ✓
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Total guesses : 12
Correct guesses : 12

A. Dharshant

ACROSS

- append method is applicable for list or tuple ✓
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Roshni M

A. Dharshant

ACROSS

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12/12
A. Dharshant

- **Reflective Critique:**

- ❖ ***Feedback of practice from students and other stakeholders:***

- The students felt that this activity used to recall information.
 - The student said this activity reduces the stress and strengthen our mind.

- ❖ ***Benefit of the practice:***

- Flipped mode of testing attracted the student's attention. They gave better focus on the topic than conventional teaching method.
 - Students got a chance to evaluate their knowledge.
 - Interaction with their class mates during the class was effective.
 - The students are able to improve analytical skills and speedup their learning capacity.

- ❖ ***Challenges faced in implementation:***

- Time consuming activity.

References:

1. <https://puzzlemaker.discoveryeducation.com/criss-cross>